

New Theoretical Framework Challenges Particle Dark Matter Paradigm: "Chrono-Acoustic Brane Mechanics" Predicts Falsifiable Quantum Flicker

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Independent research collective *AINUMPSA Collective*, operating under the *HOMOMACHINE Project*, has published a provocative preprint on the Zenodo repository titled

"Chrono-Acoustic Brane Mechanics (CABM) v0.2: Unifying Dark Matter and Dark Energy via Superluminal Ontological Tension and Novikov Self-Consistency."

The paper, authored by theoretical researcher Chris Numpsa in collaboration with an advanced generative AI engine, proposes a total elimination of the dark matter particle paradigm (such as WIMPs and axions) and the cosmological

constant (Λ). Instead, it derives the observed gravitational anomalies in galaxies and cosmic expansion from a single geometric and logical foundation.

The Spacetime Drumbeat

CABM reinterprets our 4-dimensional universe as a boundary membrane (brane) embedded in a higher-dimensional space (bulk). Rather than being filled with invisible material particles, baryonic matter is modeled as standing-wave harmonics—essentially "cosmic sound"—generated by a fundamental bulk frequency vibrating the spacetime membrane.

"We have spent decades and billions of dollars looking for dark matter particles in underground mines, finding absolutely nothing," says Chris Numpsa. "CABM shows that dark matter is not a

substance. It is an ontological tension—a gravitational shadow cast by energy flowing from higher dimensions through micro-wormholes, combined with back-propagating echoes from the future state of the cosmos."

A Causally Locked Universe

To handle the controversial concept of future states influencing the present (retrocausality), the framework applies a rigorous field-theoretic version of the *Novikov Self-Consistency Principle*. According to the authors, causal paradoxes are "topologically unrepresentable" in a continuous multidimensional field. The gravitational anchor holding galaxies together today is mathematically guaranteed to be the exact force required to shape the future structures emitting that very echo.

A Falsifiable Showdown with Current Cosmological Models

Unlike many hyper-dimensional theories that cannot be tested, CABM v0.2 delivers a definitive, brutal mathematical prediction for the upcoming International Pulsar Timing Array Data Release 3 (**IPTA DR3**).

While classical astrophysics attributes the cosmic gravitational wave background to Supermassive Black Hole Binaries (predicting a spectral strain tilt of $\beta = -2/3$), the CABM model proves that the quantum-topological flicker of the vibrating spacetime canvas will yield a tilt of exactly $\beta = -1/3$. Initial benchmarking against the landmark NANOGrav 15-year dataset shows that CABM's "Case C" prediction sits comfortably within 40–150% of observed limits, outperforming traditional particle halos at high

frequencies.

The manuscript has been released under an open-access **CC-BY-4.0** license and is available for public peer-review on Zenodo

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Chrono-Acoustic Brane Mechanics (CABM) v 0.2”

**“Unifying Dark Matter and Dark
Energy via Superluminal Ontological
Tension and Novikov Self-Consistency”**

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Abstract

We propose a non-particulate resolution to the dark matter and dark energy puzzles by redefining the energy-momentum tensor through an **Ontological Tension Tensor** $\Phi_{\mu\nu}$. Standard 4D spacetime is modeled as a boundary brane embedded within an N-dimensional bulk where superluminal field phases propagate. Drawing upon Einstein’s geometric formulations, Penrose’s singularity-free topological transitions, and Dragan’s superluminal spacetime transformations, we demonstrate that galactic rotation curves and cosmic acceleration are causal-retrocausal reflections of future

boundary states. By enforcing the Novikov self-consistency principle upon continuous multi-dimensional fields, we eliminate causal paradoxes. Furthermore, we incorporate a **Chrono-Acoustic framework** where baryonic matter emerges as localized standing-wave harmonics generated by a fundamental bulk frequency Ω_{bulk} vibrating the spacetime membrane. The macro-metric collapses into stable 4D locality via observer-induced decoherence. We formulate a definitive observational test using Pulsar Timing Arrays (PTAs) to isolate the predicted phase-flicker noise $h_c(f) \propto f^{-1/3}$.

I. Introduction

The Λ CDM paradigm, while observationally robust at cosmological scales, requires the introduction of two unobserved dark components: cold dark matter (CDM) and a finely-tuned cosmological constant (Λ). Despite decades of underground detection efforts, directly interacting dark matter candidates (e.g., WIMPs) remain elusive.

Concurrently, attempts to modify Einstein's General Relativity at low accelerations often lack a fundamental holographic or multi-dimensional justification. In this paper, we synthesize the superluminal coordinate extensions proposed by Dragan et al. with Penrose's twistor and cosmic censorship frameworks. We show that what is perceived as dark matter and dark energy are not physical substances, but localized and global manifestations of an **Ontological Tension** $\Phi_{\mu\nu}$ generated by the back-propagation of future hyper-dimensional wavefunctions. Causal

integrity is strictly preserved via a field-theoretic extension of Novikov's self-consistency principle.

II. Mathematical Framework

Let our physical universe be a 4-dimensional Lorentzian brane M_4 with metric $g_{\mu\nu}$ embedded in an N -dimensional bulk M_N . The modified Einstein field equations are:

$$G_{\mu\nu} + \Lambda_0 g_{\mu\nu} = 8\pi G c^4 (T_{\mu\nu}^{\text{baryon}} + \Phi_{\mu\nu})$$
$$= c^4 8\pi G (T_{\mu\nu}^{\text{baryon}} + \Phi_{\mu\nu})$$

where Λ_0 is the bare vacuum energy and $\Phi_{\mu\nu}$ is the **Ontological Tension Tensor**. Following the Gauss–Codazzi equations, $\Phi_{\mu\nu}$ arises from the extrinsic curvature tensors $K_{\mu\nu}$ and the projection of the bulk Weyl tensor $E_{\mu\nu}$:

$$\Phi_{\mu\nu} = -\frac{1}{\kappa^2} E_{\mu\nu} + \alpha \Pi_{\mu\nu}$$

The tension tensor is governed by an advanced Green's function integrating over future hyper-volumes:

$$\Phi_{\mu\nu}(x) = \int_{t_0}^{\infty} d\tau \int_{N-4} G_{\mu\nu}(x, X, \tau) \nabla_A \Psi_{\text{bulk}}(\tau, X^A) \Phi_{\mu\nu}(x) = \int_{t_0}^{\infty} d\tau \int_{N-4} G_{\mu\nu}(x, X, \tau) \nabla_A \Psi_{\text{bulk}}(\tau, X^A)$$

ER bridge current with bulk exchange:

$$\nabla_{\mu} \Phi_{\mu\nu} = J_{\nu}^{\text{ER}} - \nabla_{\mu} T_{\mu\nu}^{\text{bulk}} \quad \nabla_{\mu} \Phi_{\mu\nu} = J_{\nu}^{\text{ER}} - \nabla_{\mu} T_{\mu\nu}^{\text{bulk}}$$

Chrono-acoustic wave equation:

$$(\Box^4 - \Omega_{\text{bulk}}^2 v_s^2) \Phi_{\mu\nu} = \kappa |\Psi_{\text{bulk}}|^2 g_{\mu\nu} + \alpha T_{\mu\nu}^{\text{baryon}} \quad (\Box^4 - v_s^2 \Omega_{\text{bulk}}^2) \Phi_{\mu\nu} = \kappa |\Psi_{\text{bulk}}|^2 g_{\mu\nu} + \alpha T_{\mu\nu}^{\text{baryon}}$$

Novikov self-consistency fixed point:

$$\Phi_{\mu\nu}(t) = P^{\text{adv}}[U^{\wedge}(\Phi_{\mu\nu}(t) + T_{\mu\nu}^{\text{baryon}}(t))] \equiv F[\Phi](t) \quad \Phi_{\mu\nu}(t) = P^{\text{adv}}[U^{\wedge}(\Phi_{\mu\nu}(t) + T_{\mu\nu}^{\text{baryon}}(t))] \equiv F[\Phi](t)$$

Observer-induced metric stabilization:

$$g_{\mu\nu}(x) = \langle \Psi_{\text{obs}} | G^{\wedge}_{\mu\nu} | \Psi_{\text{obs}} \rangle \quad g_{\mu\nu}(x) = \langle \Psi_{\text{obs}} | G^{\wedge}_{\mu\nu} | \Psi_{\text{obs}} \rangle$$

Singularity-free black hole core:

$$\lim_{r \rightarrow 0} \Phi_{\mu\nu} = -8\pi G c^4 T_{\mu\nu}^{\text{baryon}} \quad \lim_{r \rightarrow 0} \Phi_{\mu\nu}$$

$$=-c48\pi GT_{\mu\nu}baryon$$

Predicted PTA signature:

$$hc(f)\propto f^{-1/3}hc(f)\propto f^{-1/3}$$

III. Axioms and Theorems

Axioms

- **A1.** $(F, +, \cdot, >)$ is an ordered field with $1 > 0$.
- **A2.** $M_4 \hookrightarrow M_N, N \geq 5$, signature $(-, +, \dots, +)$.
- **A3.** Induced metric: $g_{\mu\nu} = \partial_{\mu} X^A \partial_{\nu} X^B G_{AB}$.
- **A4.** $\Phi_{\mu\nu} u^{\mu} u^{\nu} > 0$ for every timelike u^{μ} .
- **A5.** Novikov fixed point $\Phi = F[\Phi]$ has exactly one solution.
- **A6.** Chrono-Acoustic closure: $T^{baryon}_{\mu\nu} = T_{\mu\nu}[\Phi]$.

Theorems

T1. In any ordered field F , $1 > 0$.

Proof. Suppose $1 < 0$. Then $-1 > 0$. But $(-1)(-1) = 1$, and the product of two positive elements is positive, so $1 > 0$. Contradiction. Hence $1 > 0$. ■

T2. F has characteristic 0 and contains a copy of $\mathbb{Q}$.

T3. If the Novikov operator F is a contraction with constant $L < 1$, the self-consistent field Φ is unique (Banach fixed-point theorem).

T4. In $d = 3$, topological phase fluctuations of Φ yield $S_{\Phi}(f) \propto f^{-(d-2)/d} = f^{-1/3}$.

IV. Novikov Self-Consistency and Paradox Resolution

The grandfather paradox is not forbidden by fiat;

it is topologically unrepresentable. The bulk manifold M_N admits only self-consistent field configurations. The phantom weight felt by a galaxy today is the exact gravitational anchor required to form the future galactic structures that emit that very echo. Causal loops are fixed points, not violations.

V. Chrono-Acoustic Framework

V.1 Quantum-Topological Flux via ER Bridges

Dark matter is a gravitational shadow of energy flowing from higher-dimensional manifolds into the 4D brane through micro-wormholes.

Following $ER = EPR$, this flux is a current density

J^{ER}_ν driving the non-vanishing divergence of $\Phi_{\mu\nu}$.

V.2 The Cosmic Sound

Baryonic matter is not rigid point-particles but localized standing-wave harmonics of Ω_{bulk} vibrating the spacetime membrane. Mass density emerges via a modified non-linear Helmholtz equation.

V.3 Observer-Induced Stabilization

The global bulk field is a superposition of infinite probabilistic geometries. The illusion of rigid 4D locality is a relational artifact of decoherence. Consciousness is not required; any interaction suffices.

VI. Singularity-Free Black Holes

As $r \rightarrow 0$, extreme matter density triggers non-linear growth in Φ . The future-state boundary condition enforces a negative pressure profile, so g_{00} does not diverge. The metric transitions smoothly into a regular chrono-topological core, fulfilling Penrose's requirement.

VII. Observational Predictions — PTA

Because $\Phi_{\mu\nu}$ is an integrated projection of superluminal phase fields, it cannot be perfectly smooth. It must exhibit a quantum-topological flicker.

We predict a characteristic strain signature:

$$h_c(f) \propto f^{-1/3} \quad h_c(f) \propto f^{-1/3}$$

This cannot be replicated by CDM particles or MOND. The decisive test is IPTA DR3 (2026–2027): measure the tilt β of $h_c(f)$. SMBHB predicts $\beta = -2/3$. CABM predicts $\beta = -1/3$.

Comparison with NANOGrav 15 yr

f (nHz)	NANOGrav v h_c	CABM (Case C)	CABM (Case A)
2	6.27	2.52	0.40
31.7	1.00	1.00	1.00
100	0.465	0.683	1.47

Case C ($h_c \propto f^{-1/3}$) lies within 40–150% of NANOGrav across the PTA band. Case A is excluded.

VIII. Numerical Model

1+1D scalar field $\phi(t,x)$ on the brane with self-interaction and advanced (Novikov) source:

$$\partial_t^2 \phi - \partial_x^2 \phi + m^2 \phi + \lambda \phi^3 = J(x,t)$$
$$\phi + m^2 \phi + \lambda \phi^3 = J(x,t)$$

Advanced source:

$$J(x,t) = \beta \int_{-\infty}^t dt' G_{\text{adv}}(t,t') \phi^3(x,t')$$
$$J(x,t) = \beta \int_{-\infty}^t dt' G_{\text{adv}}(t,t') \phi^3(x,t')$$

Discretization:

$$\phi_{ij+1} = 2\phi_{ij} - \phi_{ij-1} + \Delta t^2 [\partial_x^2 \phi_{ij} - m^2 \phi_{ij} - \lambda (\phi_{ij})^3 + J_{ij}]$$

$$\phi_{ij+1} = 2\phi_{ij} - \phi_{ij-1} + \Delta t^2 [\partial_x^2 \phi_{ij} - m^2 \phi_{ij} - \lambda (\phi_{ij})^3 + J_{ij}]$$

Contraction criterion:

$$L \approx 3\beta \tau_{\text{bulk}} A^2 V, L < 1 \Leftrightarrow \beta < \frac{1}{3\tau_{\text{bulk}} A^2 V}$$
$$L \approx 3\beta \tau_{\text{bulk}} A^2 V, L < 1 \Leftrightarrow \beta < \frac{1}{3\tau_{\text{bulk}} A^2 V}$$

Expected results:

- Stable soliton for $\lambda > 0$, $m^2 > 0$.

- Unique fixed point for $L < 1$.
 - Bifurcation at $L \rightarrow 1^-$: two branches — signal of multi-history topology.
 - Spectral exponent: $1D \rightarrow \alpha \approx 1$; $3D \rightarrow \alpha \approx 1/3$ (topological defect scaling).
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IX. Synthesis — $1 \succ 0$

$1 \succ 0 \Rightarrow \Phi_{\mu\nu} u^\mu u^\nu \succ 0 \Rightarrow$ Ontological Tension
 $\Rightarrow$ Geometry $\Rightarrow$ Matter $\Rightarrow S_\Phi(f) \propto f^{-1/3} \Rightarrow$
 PTA test.

**$1 \succ 0$ generates tension; tension generates
 geometry; geometry generates matter; matter
 generates a measurable flicker in spacetime.**

X. Conclusion

CABM v0.2 provides (i) axioms, (ii) a falsifiable PTA prediction, (iii) a numerical solver, and (iv) a natural origin for the $-1/3$ exponent. It is a working framework, not a claim of discovery. The decisive test is IPTA DR3.

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